

Gagan Jain

AI / LLM Engineer

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PROFESSIONAL SUMMARY

AI Engineer focused on production-grade GenAI systems: agentic AI orchestration, RAG pipelines, LLM fine-tuning (QLoRA/LoRA), MCP tooling, and local-first LLM deployment. Author of NexusAOS, a governance-first, event-sourced AI operating environment written in Rust (12 workspace crates, 981 tests, full CI/CD) with provider-agnostic LLM streaming (OpenAI/Anthropic), policy-enforced agent actions, and terminal/SSH integration. Strong Python + Rust foundation with end-to-end deployment: Docker, GitHub Actions, observability, and AI governance.

TECHNICAL SKILLS

AI / LLM

LLM Integration, RAG Pipelines, Agentic AI Frameworks, Prompt Engineering, Context Engineering, Multi-Agent Systems, Tool-Using Agents, Fine-tuning (QLoRA/LoRA), MCP / Tool Protocols, LangChain / LlamaIndex, Vector Databases, AI Governance

LLM OPS

Evaluation & Observability, Local LLM Deployment, OpenAI / Anthropic APIs, Embeddings, Structured Outputs / Function Calling, LiteLLM / Model Routing, Streaming Responses, NLP, Ollama

LANGUAGES

Python, TypeScript, JavaScript, Rust, C++, C, PHP, HTML/CSS

FRONTEND

React, React Native, Tailwind CSS, Bootstrap, Monaco Editor, Next.js

BACKEND

Node.js, Flask, Express, FastAPI, Axum (Rust), REST APIs, WebSockets, Laravel, Supabase

DATABASES

MySQL, MongoDB, Redis, SQLite

DEVOPS

Docker, GitHub Actions, Nginx, Gunicorn, Prometheus, Linux

VISION / ML

YOLO, OpenCV, Computer Vision

PROJECTS

NexusAOS AI/AGENTIC

Governance-first, event-sourced AI operating environment in Rust — 12 workspace crates, 981 tests, provider-agnostic LLM streaming.

- Designed a governance-first, event-sourced AI OS in Rust: 12 workspace crates, 981 passing tests, 0 clippy warnings, full GitHub Actions CI/CD
- Built a policy-enforced agent kernel where LLMs propose actions and the kernel validates and records every state change in an append-only audit trail
- Implemented a provider-agnostic LLM layer with OpenAI-compatible and Anthropic streaming, LiteLLM routing, and local-first inference that runs fully offline
- Delivered native terminal emulation (PTY + VT100), SSH multiplexing, a secrets vault, and four interfaces: CLI, TUI, GUI, RPC



Source Code



Documentation

SeshaOS AI/AGENTIC

NexusAOS v2 — governance-first, local-first AI OS with specialist local models and LiteLLM-compatible routing.

- Architected specialist model stack: Gemma 4 12B (Planner), Qwen3-Coder 30B (Implementation), Qwen3.5 9B (Vision)
- Designed kernel-centric governance where models propose actions and the kernel validates/records every state change
- Built a LiteLLM-compatible proxy for NVIDIA NIM and other model providers
- Maintained event-sourced architecture with reversible, permissioned actions and offline-first execution



Source Code



Documentation

Vyākṛti FLAGSHIP

Sanskrit-oriented programming language with complete compiler pipeline and browser-based IDE. 123 tests.

- Complete compiler pipeline: lexer → parser → type checker → bytecode compiler — all built from scratch in Rust
- Browser-based IDE with React, Monaco Editor, syntax highlighting, autocomplete, and diagnostics
- Rust (axum) backend with compile, REPL, LSP, and file management endpoints via REST + WebSocket
- 123 tests covering the full pipeline, including a self-hosting corpus

 [Source Code](#)

 [Documentation](#)

AIM — Attendance Information Manager PRODUCTION-READY

Production-grade Flask + MySQL platform with Argon2id auth, Prometheus monitoring, and CI/CD.

- Production-grade Flask + MySQL platform with Argon2id auth, CSRF protection, brute-force lockout, breached-password scanning, JWT sessions, and strict CSP/HSTS headers
- Deployed Prometheus metrics, structured JSON logging, Chart.js analytics, FullCalendar scheduling, Docker Compose, and GitHub Actions CI/CD; 101 automated pytest tests

 [Source Code](#)

 [Documentation](#)

RAG Service — Production RAG + Vector Search GEN AI

Production RAG API service: hybrid retrieval (vector + keyword), embeddings, reranking-ready, and an LLM-as-judge eval harness.

- Built a FastAPI RAG service with hybrid retrieval (dense embeddings + BM25 keyword, merged via Reciprocal Rank Fusion) over ChromaDB
- Implemented chunking strategies (recursive, semantic), embedding pipelines, and an /ask endpoint that streams grounded answers with citations
- Shipped LLM-as-judge evaluation harness (faithfulness, answer relevance, correctness) with golden-set CI gate and RAGAS-style reports
- Containerized with Docker Compose; OpenAI-compatible LLM interface with local-first fallback

 [Source Code](#)

 [Documentation](#)

LLM Eval Harness GEN AI

Reusable LLM evaluation harness: golden-set YAML, LLM-as-judge metrics, offline heuristic fallbacks, and CI-ready reports.

- Built a golden-set-driven evaluation harness with faithfulness, answer-relevance, and correctness metrics (LLM-as-judge + lexical fallbacks)
- Added offline heuristic scorers so evals run in CI without API keys; JSON + Markdown reports for regression tracking
- Designed golden-set YAML format (question, golden answer, context) and a CLI: `python -m eval_harness --golden-set ...`
- Containerized and CI-ready; used to gate the RAG service pipeline

 [Source Code](#)

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EXPERIENCE

Industry 5.0 Industrial Automation Trainee

CodenPlay Robotics

VIT Vellore / Online

Aug – Nov 2025

- Built and debugged PLC automation logic (ladder logic, structured text) in CODESYS
- Simulated industrial process workflows in Factory I/O
- Implemented flow-based control/integration in Node-RED
- Covered HMI design, industrial communication, process monitoring, and the debugging→testing→deployment workflow

EDUCATION

VIT Vellore

B.Tech in Computer Science and Engineering

Graduated 2025